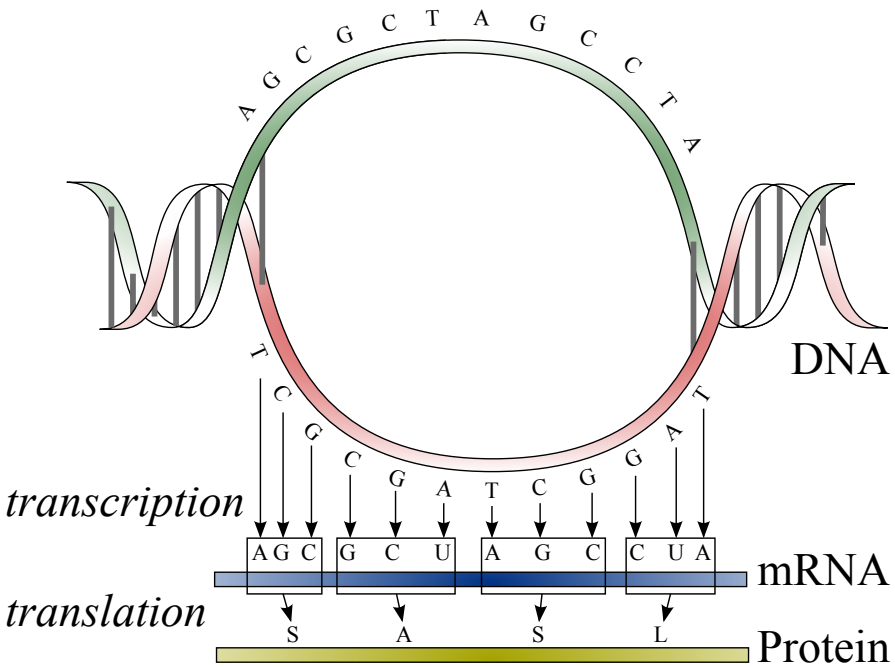
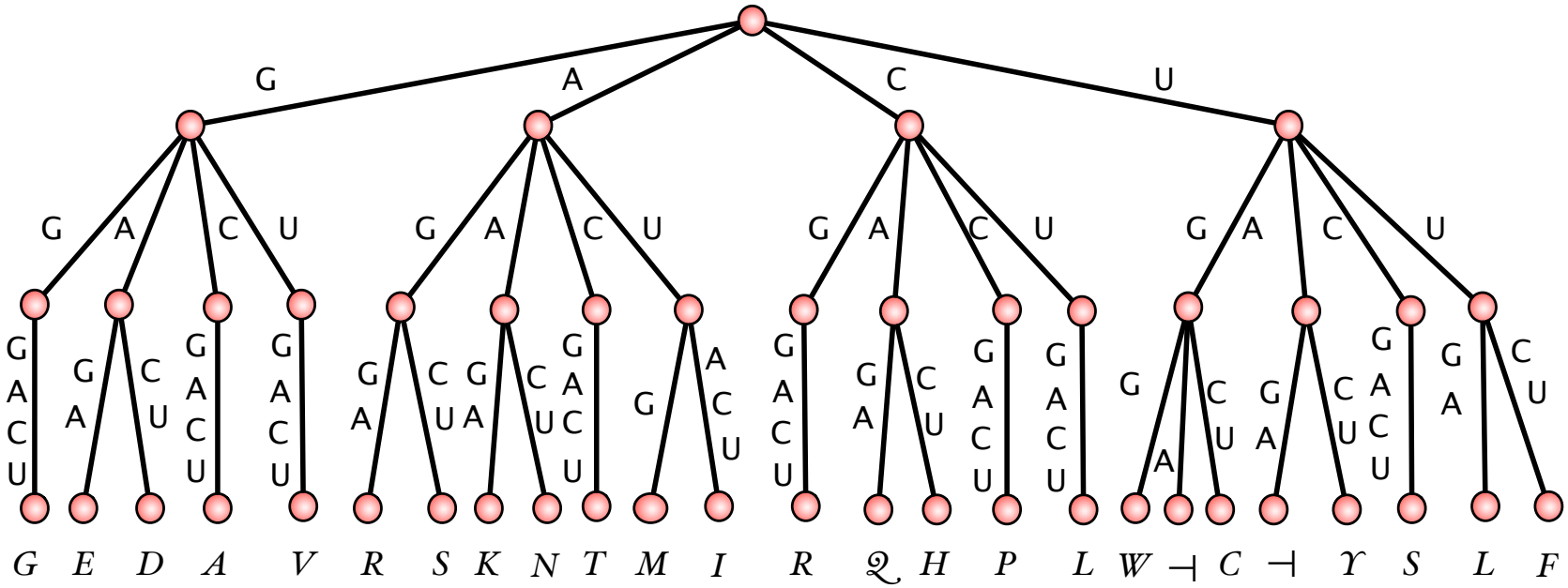
Protein Structure Prediction

Proteins and Central Dogma



- The translation phase starts from a mRNA sequence and associates a protein sequence
- Proteins are made of amino acids (20 common different types)
- Amino acids are defined by letters $\{A, \ldots, Z\} \setminus \{B, J, O, U, X, Z\}$

Universal code

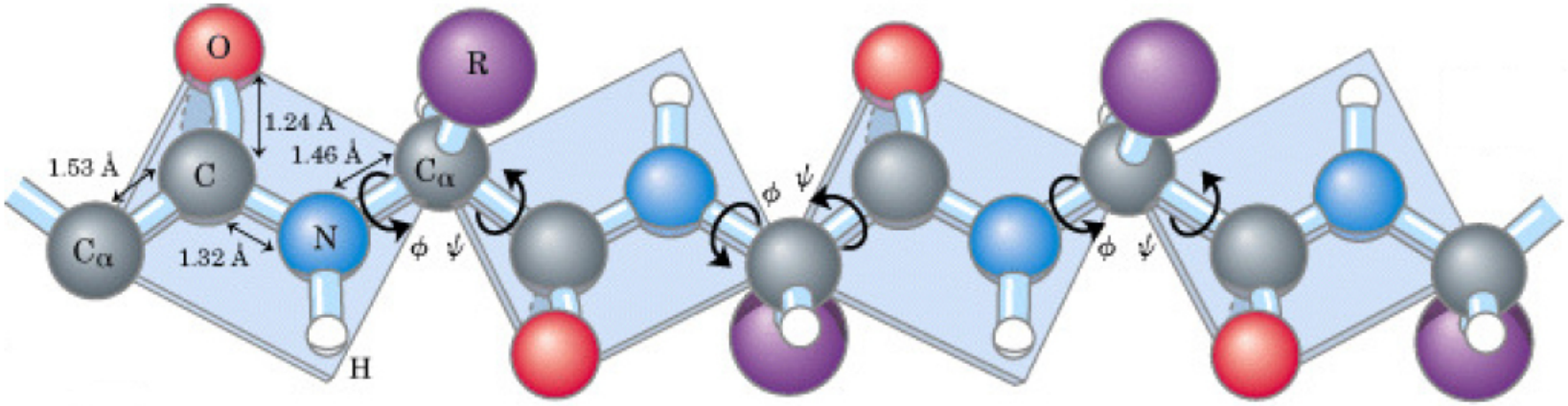


- The translation selects 3 RNA basis and associates 1 amino acid.
- The translation rules are encoded in the universal code.
- The code contains *stop* symbol and some redundant RNA triplets.

Proteins

Amino acids

- Proteins are molecules made of a linear sequence of amino acids.
- Amino acids are combined through *peptide bond*.



- The purple dots represent the *side chains*, that depend on the amino acid type
- Side chains have different shape, size, charge, polarity, etc.
- A side chain contains from 1 (Glycine) up to 18 (Tryptophan) atoms.

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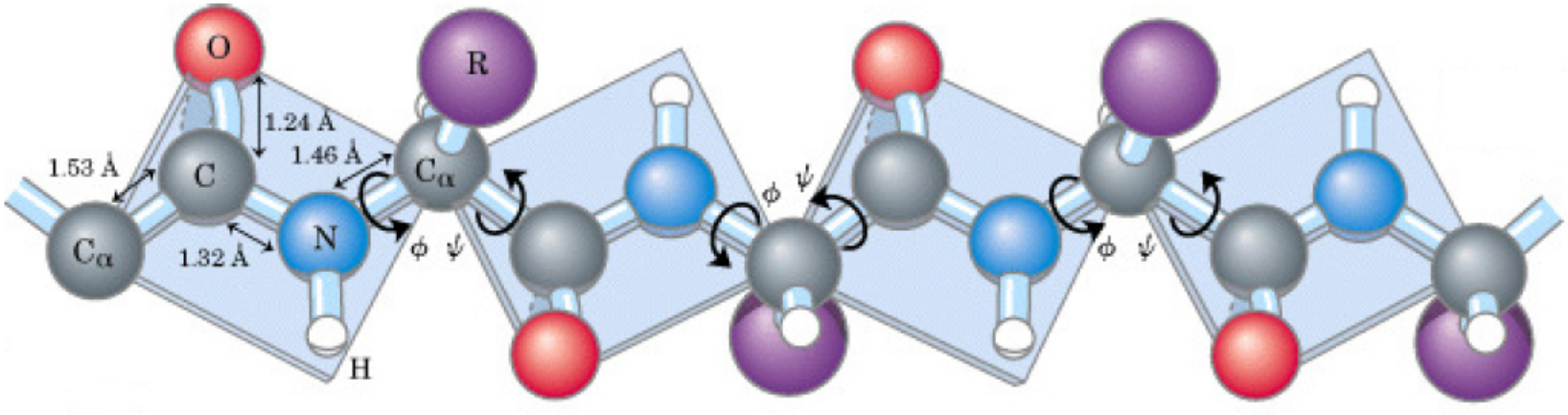

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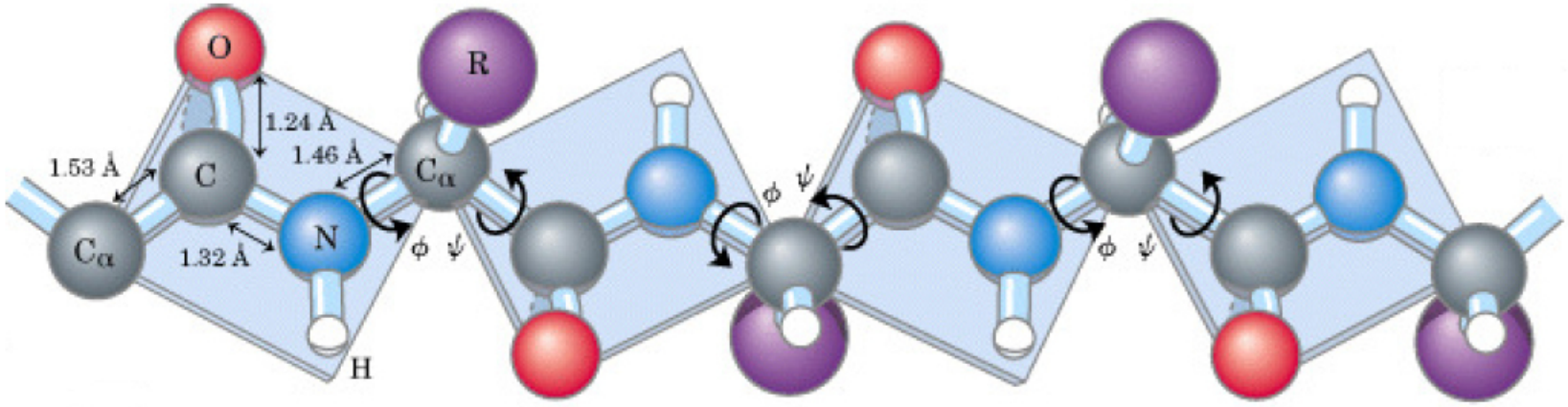
Proteins

Amino acids



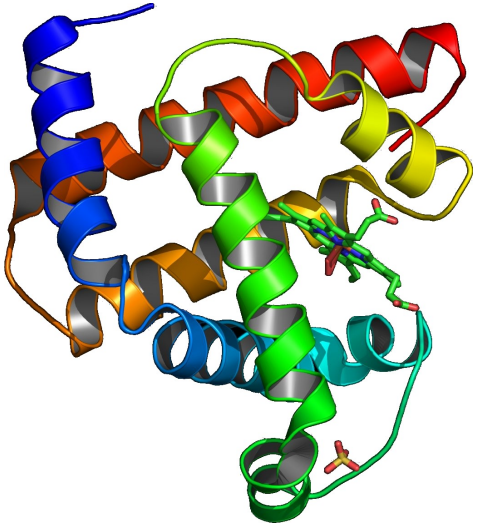
- There are 2 degrees of freedom (black arrows) for each amino acid
- A protein with n amino acids has $2n$ degrees of freedom!
- Typical size range from 50 to 500 amino acids

The structure prediction problem

- Given the primary structure of a protein (its amino acid sequence)
- For each amino acid, output its position in the space (tertiary structure of a protein)

A	L	F	W	K	L	R	R	...

? ⇓ ?



Proteins

Facts

- Proteins FOLD spontaneously (when in their natural environment)
- Folding is consistent $\Rightarrow$ same protein folds in the same way [Anfinsen74]
- Folding is fast $\Rightarrow$ 1mS – 1S
- Driven by non covalent forces: electrostatic interactions, volume constraints, Hydrogen Bonding, van der Waals, Salt/disulfide Bridges
- Backbone is rigid, interaction with water, ions and ligands

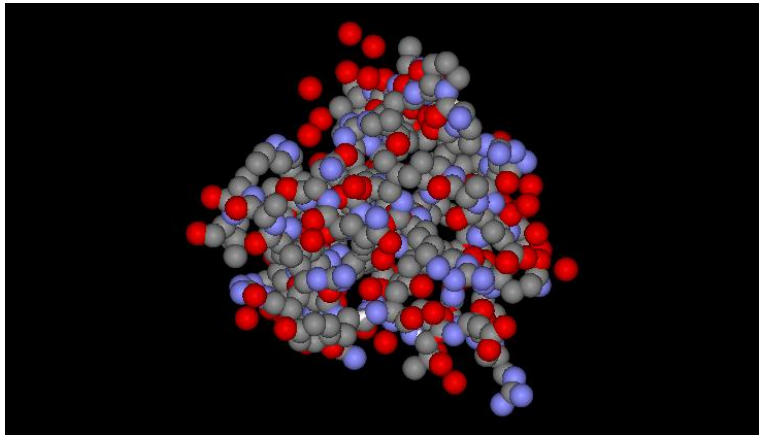
The structure prediction problem

... and this is the hard part:

- In nature a protein has a unique/stable 3D conformation
- A cost function (that mimics physics laws) can be used to score each conformation
- Searching for the optimal score produces the best candidate is difficult (NP-complete even in extremely simplified modelings)

The protein structure prediction problem

- In this presentation we have two simplifications:
- Protein model: only one atom per amino acid, only 2 classes of amino acids (hydrophobic and polar)



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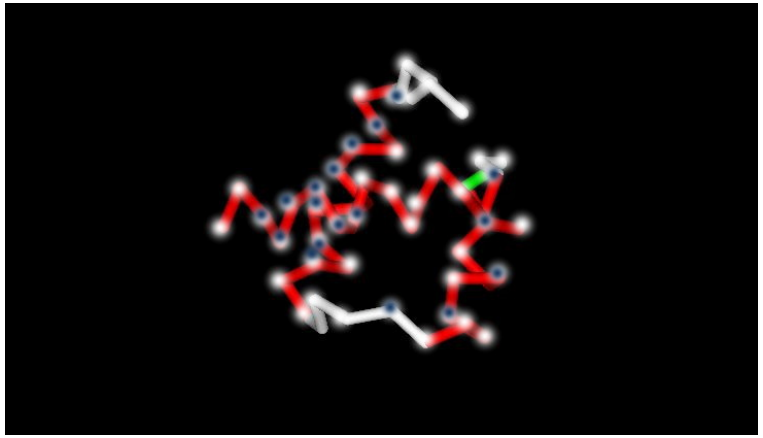

- Spatial model: 2D square lattice to represent amino acid positions

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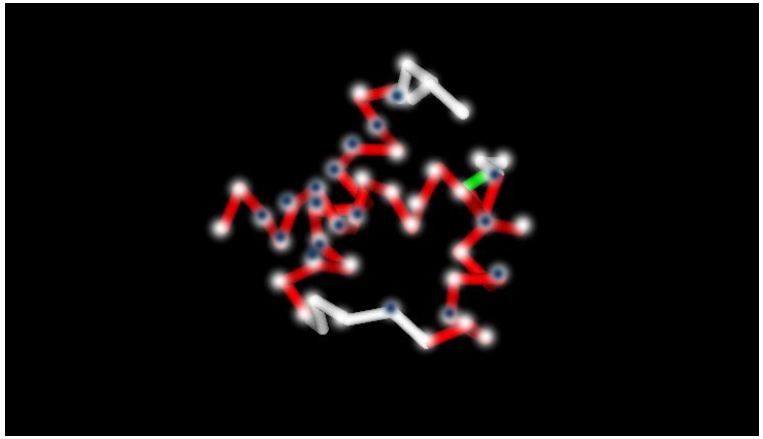

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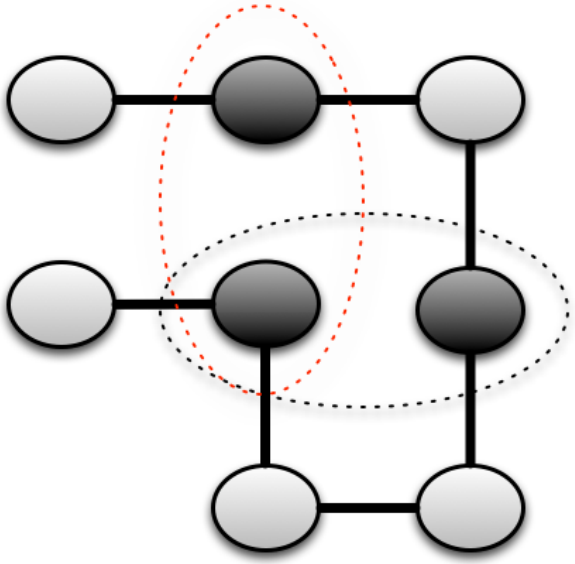
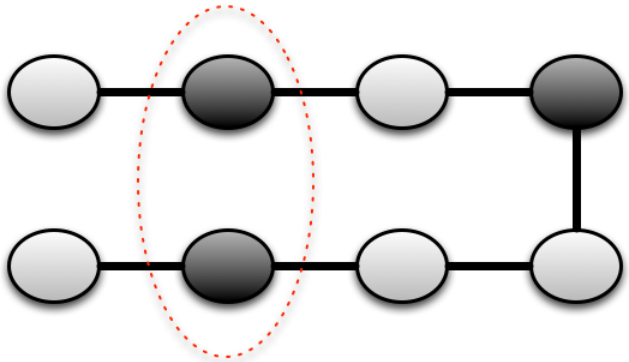
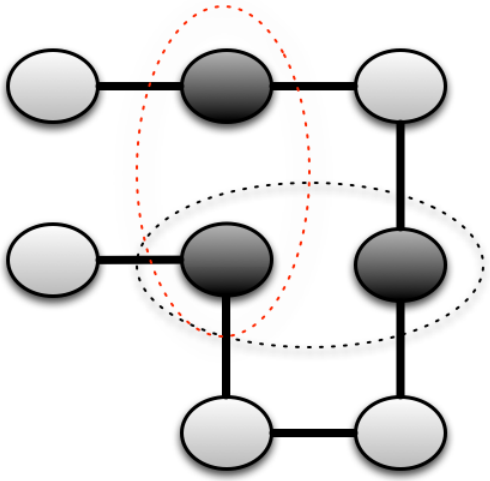


The protein structure prediction problem

Model

- The input is a list S of amino acids $S = s_1, \ldots, s_n$,
- where $s_i \in \{h, p\}$
- Each s_i is placed on a 2D grid with integer coordinates
- Any pair of two amino acids can't occupy the same position
- If two amino acids are at distance 1, they are in contact





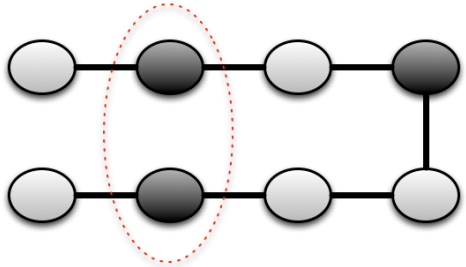
The protein structure prediction problem

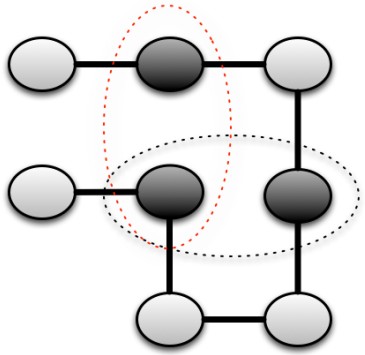
Model

- A folding is a function $\omega : \{1, \ldots, n\} \longrightarrow \mathbb{N}^2$ where
- $\forall i \, \text{next}(\omega(i), \omega(i+1))$ and
- $\forall i, j \, (i \neq j \rightarrow \omega(i) \neq \omega(j))$
- $\text{next}(\langle X_1, Y_1 \rangle, \langle X_2, Y_2 \rangle) \Longleftrightarrow |X_1 - X_2| + |Y_1 - Y_2| = 1.$
- Find a folding that minimizes the (simplified) energy function:

$$E(S, \omega) = \sum_{\substack{1 \leq i \leq n-2 \\ i+2 \leq j \leq n}} \text{Pot}(s_i, s_j) \cdot \text{next}(\omega(i), \omega(j))$$

where $\text{Pot}(p, p) = \text{Pot}(h, p) = \text{Pot}(p, h) = 0$ and $\text{Pot}(h, h) = -1$.





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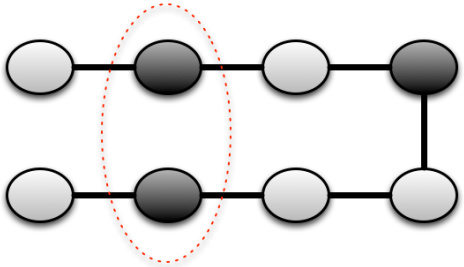

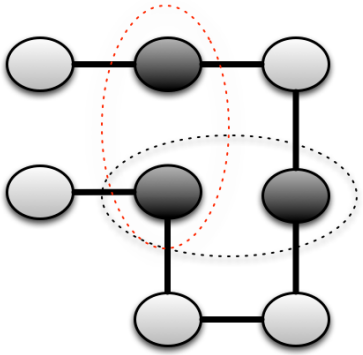

- With $\mathbb{N}^2$ and HP, establishing whether there is a folding with energy $< k$ is NP-complete
- (Crescenzi, Goldman, Papadimitriou, Piccolboni, Yannakakis. On the Complexity of Protein Folding. Journal of Computational Biology 5(3): 423-466 (1998))

References

- Alessandro Dal Palù, Agostino Dovier, Andrea Formisano, Enrico Pontelli: ASP Applications in Bio-informatics: A Short Tour. Künstliche Intelligenz 32(2-3): 157-164 (2018)
- Alessandro Dal Palù, Agostino Dovier, Andrea Formisano, and Enrico Pontelli. Exploring Life: Answer Set Programming in Bioinformatics. Chapter 7 of Declarative Logic Programming, Theory, Systems, and Applications. ACM Press, New York, NY, USA 2018, pp. 359-426.